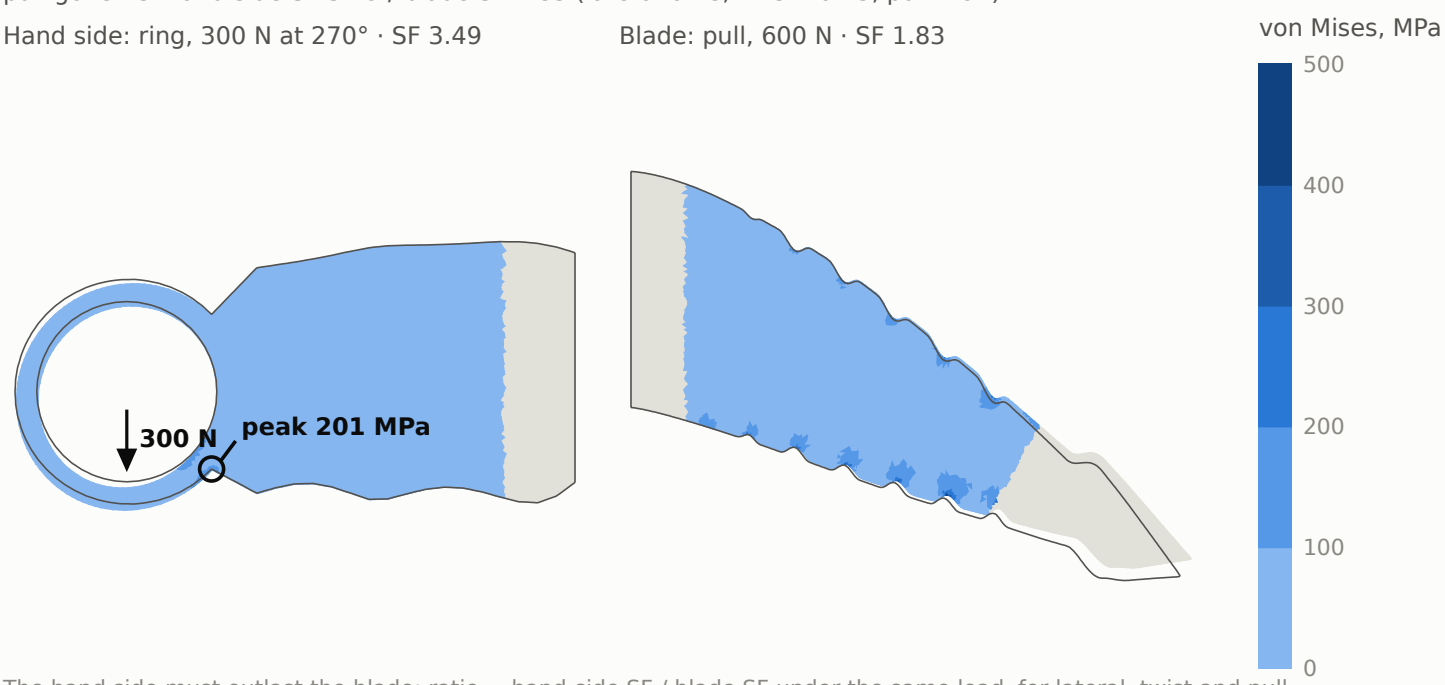
Blade before hand (failure sequence) · ratio 1.911 (required 1.2) · margin 1.592

pull governs: hand side SF 3.49 / blade SF 1.83 (lateral 6.18, twist 10.75, pull 1.91)

Hand side: ring, 300 N at 270° · SF 3.49

Blade: pull, 600 N · SF 1.83



The hand side must outlast the blade: ratio = hand-side SF / blade SF under the same load, for lateral, twist and pull. One colour scale; grey: not read; deformation ×10 (ring), ×10 (blade). Hand-side pull SF = min(grip formula, ring). Reproduces the record's SF bit for bit (Tier 2 at h 0.2, same platform).